

Fall 2026 CEC 470 Computer Architecture

Homework 1

Total: 100 points

Due Date: 9/13/2026

1. Define the five classic components of a computer. (5 points)
2. Programs and data are both stored in memory in a Von Neumann machine. Discuss **one advantage** and **one disadvantage** of this design. (10 points)
3. Machine code instructions are stored as bit sequences. If an instruction is 32 bits wide, how many unique instructions can be represented? (5 points)
4. What is the difference between **computer architecture** and **computer organization**? (10 points)
5. Give an example of a situation where two machines share the same ISA but have different organizations. (5 points)

- List three important components included in an ISA. (3 points)
- ARM and x86 are two widely used ISAs. Research and compare **one advantage** and **one disadvantage** of each in terms of performance, power consumption, or compatibility. (25 points)
- A program runs in 18 seconds on Computer A with a 600 MHz clock. Computer B's performance (in terms of execution time) is 2 times better than Computer A. Given that Computer requires $1.5\times$ more cycles per program, what clock rate must Computer B target? (Show your work) (15 points)

9. A program has the following instruction mix:

40% Class A (1 cycle),

40% Class B (2 cycles),

20% Class C (3 cycles).

(a) Compute the **average CPI**. (5 points)

(b) If the machine's clock cycle time is 1 ns, calculate the total execution time for 1 million instructions. (7 points)

10. A program spends 80% of its time performing floating-point operations. If floating-point performance is improved by a factor of 10, what is the maximum possible speedup? (10 points)